

- This exam contains 15 pages (including the front and back of this cover page) and 16 problems. Check to see if any pages are missing.
- This is the third exam and will count for 20% of your final grade. You have 120 minutes (5:00 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 90 points total in this exam, as well as 1 bonus questions worth a total of 3 possible points.

Page:	3	4	5	6	8	9	10	11	12	13	Total
Points:	12	10	10	15	10	5	10	5	5	8	90
Score:											

Bonus:	
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1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

☐ 2 pts

(a) $\sec(x) \csc(x) \cot(x) = \csc^2(x)$.

☐ True☐ False☐ 2 pts

(b) $\frac{\sec(x)}{\tan(x)} = \cos(x)$.

☐ True☐ False☐ 2 pts

(c) $\tan\left(\frac{3\pi}{4}\right) = \frac{-1 + \sqrt{3}}{1 + \sqrt{3}}$

☐ True☐ False☐ 2 pts

- (d) The absolute value of the complex number $-5.6 - 6.8i$ is approximately 8.81.

☐ True☐ False☐ 2 pts

(e) In polar form, $6i = 6\text{cis}\left(\frac{\pi}{2}\right)$

☐ True☐ False☐ 2 pts

(f) $\frac{\cot(\theta)}{\tan(\theta)} = \csc^2(\theta) \sec^2(\theta)$

☐ True☐ False

5 pts

2. Simplify the expression

$$\frac{\tan\left(\frac{4}{7}x\right) + \tan\left(\frac{3}{7}x\right)}{1 - \tan\left(\frac{4}{7}x\right)\tan\left(\frac{3}{7}x\right)}$$

☐ (A) $\tan(x)$

☐ (B) $\tan(7x)$

☐ (C) $\tan(3x)$

☐ (D) $\tan(-4x)$

5 pts

3. Find the exact value of $\tan\left(\frac{13\pi}{12}\right)$.

☐ (A) -1

☐ (B) $\frac{1}{2}$

☐ (C) 1

☐ (D) $2 - \sqrt{3}$

5 pts

4. Choose the option below that is equal to $4 \sin(7x) \cos(7x)$.

☐ Ⓐ $2 \sin(14x)$

☐ Ⓑ $2 \cos(14x)$

☐ Ⓒ $\sin\left(\frac{7}{2}x\right)$

☐ Ⓓ $\sin(3x) - \sin(4x)$

5 pts

5. Choose the option below that is equal to $10 \sin(9x) \sin(5x)$.

☐ Ⓐ $5 [\sin(14x) - \cos(4x)]$

☐ Ⓑ $5 [\cos(4x) - \cos(14x)]$

☐ Ⓒ $10 \cos(2x) \cos(7x)$

☐ Ⓓ $20 \sin(7x) \cos(2x)$

5 pts

6. Simplify the expression to one term:

$$\sin(x) \cos(10x) - \cos(x) \sin(10x).$$

$$\textcircled{\text{A}} \tan\left(\frac{x}{10}\right)$$

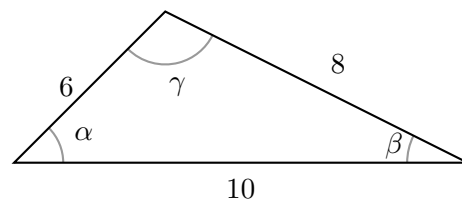
$$\textcircled{\text{B}} \cos(11x)$$

$$\textcircled{\text{C}} -\sin(9x)$$

$$\textcircled{\text{D}} \cos(9x)$$

10 pts

7. Find the area of the triangle below. Round your answer to the nearest tenth.



$$\textcircled{\text{A}} 31.9$$

$$\textcircled{\text{B}} 281.3$$

$$\textcircled{\text{C}} 303.6$$

$$\textcircled{\text{D}} 24$$

8. Solve the equation exactly on the interval $0 \leq \theta < 2\pi$.

$$\frac{\sin(2\theta)}{\sqrt{3}\sin(\theta)} = 1$$

- Ⓐ $\theta = \frac{\pi}{12}, \theta = \frac{11\pi}{12}, \theta = \frac{13\pi}{12},$ and $\theta = \frac{23\pi}{12}$
Ⓑ $\theta = \frac{\pi}{6}, \theta = \frac{11\pi}{6}, \theta = \frac{13\pi}{6},$ and $\theta = \frac{23\pi}{6}$
Ⓒ $\theta = \frac{\pi}{24}, \theta = \frac{11\pi}{24}, \theta = \frac{13\pi}{24},$ and $\theta = \frac{23\pi}{24}$
Ⓓ $\theta = \frac{\pi}{3}, \theta = \frac{11\pi}{3}, \theta = \frac{13\pi}{3},$ and $\theta = \frac{23\pi}{3}$

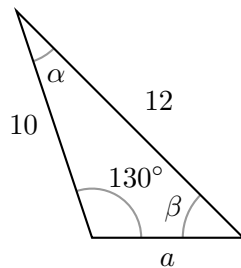
10 pts

9. Solve the equation exactly on the interval $[0, 2\pi)$.

$$\sin^2(x)(1 - \cos^2(x)) - \cos^2(x)(1 - \sin^2(x)) = 0$$

- Ⓐ $x = \frac{\pi}{12}, x = \frac{\pi}{4}, x = \frac{5\pi}{12},$ and $x = \frac{7\pi}{12}.$
- Ⓑ $x = \frac{\pi}{8}, x = \frac{3\pi}{8}, x = \frac{5\pi}{8},$ and $x = \frac{7\pi}{8}.$
- Ⓒ $x = \frac{\pi}{4}, x = \frac{3\pi}{4}, x = \frac{5\pi}{4},$ **and** $x = \frac{7\pi}{4}.$
- Ⓓ $x = \frac{\pi}{2}, x = \frac{3\pi}{2}, x = \frac{5\pi}{2},$ and $x = \frac{7\pi}{2}.$

5 pts

10. Solve for a in the triangle below. Round your answer to the nearest tenth.

Ⓐ 2.2

Ⓑ 4.1

Ⓒ 2.8

Ⓓ 3.1

5 pts

11. Convert the point $(2\sqrt{2}, 2\sqrt{2})$ from rectangular coordinates into polar coordinates (r, θ) with $0 \leq r$ and $0 \leq \theta < 2\pi$.

Ⓐ $\left(2\sqrt{2}, \frac{3\pi}{4}\right)$

Ⓑ $\left(8, \frac{\pi}{4}\right)$

Ⓒ $\left(4, \frac{\pi}{4}\right)$

Ⓓ $\left(4, \frac{5\pi}{4}\right)$

5 pts

12. Let z_1 and z_2 be complex numbers given in polar form by

$$z_1 = 25\text{cis}\left(\frac{13\pi}{12}\right), \text{ and } z_2 = 5\text{cis}\left(\frac{\pi}{12}\right).$$

Find $\frac{z_1}{z_2}$ in polar form.

Ⓐ $5\text{cis}\left(\frac{3\pi}{4}\right)$

Ⓑ $\sqrt{5}\text{cis}\left(\frac{3\pi}{4}\right)$

Ⓒ $5\text{cis}\left(\frac{7\pi}{6}\right)$

Ⓓ $\frac{1}{5}\text{cis}\left(\frac{3\pi}{4}\right)$

5 pts

13. Solve the system by any method.

$$\begin{cases} 5x - 5y = 5/4 \\ 4x - 5y = 1 \end{cases}$$

$$x = \underline{\frac{1}{4}},$$

$$y = \underline{0}$$

$$x = \underline{\hspace{2cm}},$$

$$y = \underline{\hspace{2cm}}$$

5 pts

14. The admission fee at a small fair is \$1.50 for children and \$4.00 for adults. On a certain day, 2200 people enter the fair and \$5050 is collected. How many children and how many adults attended?

Adults: 700,Children: 1500

Adults: _____,

Children: _____

8 pts

15. Solve the system by any method.

$$\begin{cases} x - 2y + 3z &= 7 \\ 2x + y + z &= 4 \\ -3x + 2y - 2z &= -10 \end{cases}$$

$x = \underline{\textcolor{red}{2}}$

$y = \underline{\textcolor{red}{-1}}$

$z = \underline{\textcolor{red}{1}}$

$x = \underline{\hspace{2cm}},$

$y = \underline{\hspace{2cm}}$

$z = \underline{\hspace{2cm}}$

3 (bonus) 16. Solve the system by any method.

$$\begin{cases} y &= x^2 - 2x + 8 \\ y &= 5x - 2 \end{cases}$$